

STOCHASTIC MAGNUS-BESSEL TENSOR-TRAIN SELF-CONSISTENCY BRIDGES FOR ELLIPTIC AND HYPOELLIPTIC DIFFUSIONS

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ABSTRACT

Learning neural diffusion bridges that condition stochastic differential equations (SDEs) onto specified terminal states is fundamental to rare-event simulation, transition path sampling, and generative modeling. Existing self-consistency drift-control frameworks suffer from four coupled failure modes: single-step backward target recursions inject $\mathcal{O}(1/\delta t)$ Brownian variance and explode on steep potential barriers; near-terminal score guidance diverges through a $(T - t)^{-1}$ singularity; unconstrained drift parameterizations introduce spurious rotational components; and direct drift inversion breaks down entirely on hypoelliptic kinetic SDEs with singular diffusion. We present the **Stochastic Magnus-Bessel Tensor-Train Self-Consistency Bridge (SMBTT-CCDB)**, a unified framework that resolves all four. SMBTT-CCDB contributes: (1) a non-recursive Lie-algebraic Stochastic Magnus Expansion target cumulant operator combined with a fixed auxiliary change of measure ($\tilde{b} = -b$) that removes base-drift Jacobian compounding while retaining state-dependent diffusion cumulants; (2) a velocity-capped manifold score guide that couples a floored temporal metric with a saturating cap to eliminate boundary singularities; (3) exact irrotational ConservativeMLP and low-rank Tensor-Train B-spline potentials that guarantee curl-free control ($\sim 10^{-15}$ relative curl); and (4) a Cotangent Bundle Poisson Projection that maps control targets strictly into actuated velocity channels. Across seven elliptic benchmarks, SMBTT-CCDB matches or outperforms self-consistency baselines with a single fixed configuration, while training up to $4.9\times$ faster than trajectory-differentiating methods. On hypoelliptic kinetic Langevin SDEs where standard methods fail (infinite KL), SMBTT-CCDB attains exact kinetic-constraint preservation ($dX_t = V_t dt$) with zero position-drift error and finite velocity KL.

1 INTRODUCTION

Diffusion bridges provide a rigorous framework for conditioning continuous stochastic differential equations (SDEs) to hit specified terminal target sets, states, or distributions within a finite horizon T (Vargas et al., 2021; Nodozi et al., 2023; Pariset et al., 2023). Sampling conditioned diffusion paths is central to scientific machine learning: rare-event simulation, transition path sampling in molecular dynamics, interest-rate modeling, and generative modeling (Cheng et al., 2017; Howard et al., 2025; Yang et al., 2025). Casting bridge estimation as a stochastic optimal control (SOC) problem, recent online neural methods optimize a drift-modification field $u(t, X_t)$ that forces the base dynamics toward the target while minimizing kinetic energy (Domingo-Enrich et al., 2024; Domingo-Enrich & Han, 2026; Shin et al., 2026).

Despite this progress, leading neural bridge frameworks—most notably Control Consistency Diffusion Bridges (CCDB) (Howard et al., 2025) and Neural Guided Diffusion Bridges (NGDB) (Yang et al., 2025)—share four structural vulnerabilities that limit their use on realistic physical systems:

1. **High-variance backward recursions and Jacobian explosions.** Self-consistency training builds regression targets U_s recursively backward by multiplying by discrete single-step Jacobians $J_{s+\delta t|s} \approx I + \nabla b \delta t + \nabla \sigma \delta B_s$. Injecting the Brownian increment δB_s causes $\mathcal{O}(1/\delta t)$ target

variance, and compounding $(I + \nabla b \delta t)$ across steep barriers where ∇b has positive eigenvalues triggers gradient explosions (Howard et al., 2025; Tang et al., 2024).

2. **Terminal boundary singularities.** Guiding flows toward x_T typically introduce $(T - t)^{-1}$ velocity singularities as $t \rightarrow T$, destabilizing near-terminal training (Shin et al., 2026; Zhou et al., 2023; Chen et al., 2021).
3. **Rotational drift errors.** Self-consistency guarantees recovery of the unique bridge control only when u_θ is of gradient form ($u = \nabla_x \psi$) (Howard et al., 2025). Arbitrary vector-field parameterizations carry non-zero curl, violating the fixed-point guarantee (Paszke et al., 2019).
4. **Breakdown on hypoelliptic (kinetic) SDEs.** For kinetic Langevin systems $dX_t = V_t dt$, $dV_t = b_v dt + \sigma_v dB_t$, noise acts only on velocity, so $\sigma\sigma^\top$ is singular and direct inversion $(\sigma\sigma^\top)^{-1}(f - b)$ fails. Naive inversion injects drift into the un-actuated position channel ($dX_t \neq V_t dt$), yielding physically invalid paths and infinite KL (Cheng et al., 2017; Ditlevsen & Samson, 2018; Dockhorn et al., 2021; Holdijk et al., 2022).

We introduce **SMBTT-CCDB**, a unified framework for robust neural diffusion bridge estimation across elliptic and hypoelliptic dynamics (Fig. 1). Its four contributions map one-to-one onto the failures above:

- **Continuous-time Stochastic Magnus target operator.** We replace noisy discrete recursions with a Lie-algebraic Stochastic Magnus Expansion (SME) path-integral cumulant operator $\mathcal{M}_{s,t}(X)$. Combined with an auxiliary change of measure $\tilde{b} = -b$ (SC2), the base-drift Jacobian ∇b is removed from multiplicative growth, preventing explosions while retaining exact diffusion cumulants $\nabla \sigma$.
- **Velocity-capped non-singular score guidance.** We reformulate manifold score guidance with a floored metric $\max(T - t, h_{\text{floor}})$ and a saturating cap $v_{\text{max}} \tanh(\|f\|/v_{\text{max}})$, strictly bounding near-terminal velocity while enabling conditioning on points, sub-manifolds, or target point clouds.
- **Exact structural irrotationality.** We parameterize $u_\theta = \nabla_x \psi_\theta$ (ConservativeMLP for $d \leq 50$; low-rank Tensor-Train B-spline potentials for $d \geq 50$), guaranteeing curl-free control to machine precision.
- **Cotangent Bundle Poisson Projection.** For singular $\sigma\sigma^\top$, we project targets exclusively into actuated velocity channels, enforcing $dX_t = V_t dt$ with exact zero position-drift error—the first self-consistency solver for underdamped Langevin bridges.

Empirically, SMBTT-CCDB matches or beats CCDB and competitive baselines across seven elliptic benchmarks (CIR, Double Well, Cell Differentiation, and 2D/100D Müller-Brown) with a single fixed setup, and uniquely solves hypoelliptic kinetic Langevin bridges.

2 RELATED WORK

Controlled neural bridges and adjoint matching. Conditioning diffusions between fixed endpoints was classically tackled by MCMC/SMC (Vargas et al., 2021; Seong et al., 2024). Recent work frames bridge simulation as SOC, learning drift controls online along simulated paths (Domingo-Enrich et al., 2023; Domingo-Enrich, 2024). Adjoint Matching methods (Domingo-Enrich et al., 2024; Domingo-Enrich & Han, 2026; Shin et al., 2026; Liu et al., 2025; So et al., 2026; Choi et al., 2025; Park et al., 2025) compute drift adjustments via backward adjoint SDEs, and NGDB (Yang et al., 2025) backpropagates through trajectories; both incur $\mathcal{O}(T)$ memory. CCDB (Howard et al., 2025) removes trajectory backpropagation using a fixed-point self-consistency objective, related to self-consistency and score-matching principles for operator learning (Yang et al., 2024) and Schrödinger Bridge Matching (Shi et al., 2023). CCDB, however, relies on discrete single-step recursions that amplify δB_s and explode on barriers (Howard et al., 2025; Ditlevsen & Samson, 2018); SMBTT-CCDB replaces them with a continuous SME cumulant plus $\tilde{b} = -b$.

Simulation-free bridge estimators. An alternative avoids online controlled simulation, training on uncontrolled reference paths. Diffusion Schrödinger Bridges (Bortoli et al., 2021b) solve entropy-regularized OT via IPF and score matching (Bortoli et al., 2021a; Chen et al., 2021; Tang et al., 2024); simulation-free variants (Tong et al., 2023) and DDBM (Zhou et al., 2023) use static OT couplings or flow matching. Doob’s Lagrangian (Du et al., 2024) and DBVI (Xu et al., 2025) parameterize path

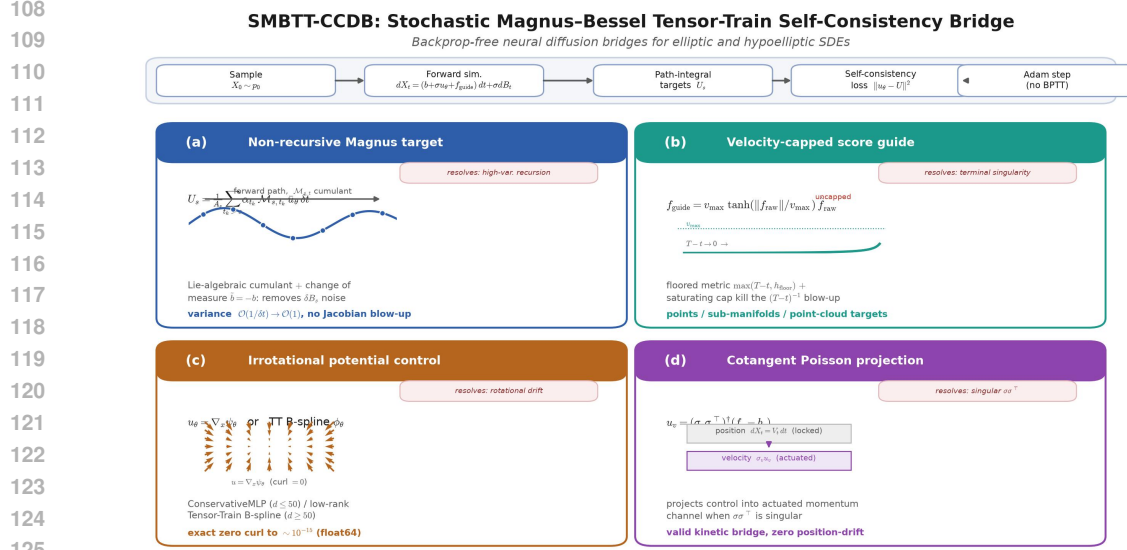


Figure 1: Overview of the SMBTT-CCDB framework. **(a)** Non-recursive target evaluation via a pathwise Stochastic Magnus cumulant operator with the $\tilde{b} = -b$ change of measure. **(b)** Velocity-capped manifold score guidance removing the terminal drift singularity. **(c)** Conservative potential control (ConservativeMLP / Tensor-Train B-spline) with exact zero curl. **(d)** Cotangent Bundle Poisson Projection mapping control into actuated velocity channels for hypoelliptic diffusions.

measures with Gaussian ansatzes, while energy-based models enforce Fokker-Planck consistency (Plainer et al., 2025). These degrade sharply over high energy barriers where base dynamics rarely reach the target (Howard et al., 2025; Yang et al., 2025; Du et al., 2024); SMBTT-CCDB instead uses online controlled simulation with velocity-capped guidance.

Hypoelliptic and manifold-constrained SOC. In underdamped Langevin and mechanical systems, noise acts on a restricted subspace (Cheng et al., 2017; Arnaudon et al., 2020; Bortoli et al., 2022; Holdijk et al., 2022), giving hypoelliptic SDEs under Hörmander’s condition (Habermann, 2018; Guillin & Monmarché, 2016) with singular $\sigma\sigma^\top$. Critically-damped score-based models (Dockhorn et al., 2021; Shi & Liu, 2024) and horizontal/Hamiltonian score matching (Grong et al., 2024; Holderrieth et al., 2024; Baker et al., 2024) circumvent singularity via velocity conditionals, and momentum Schrödinger bridges—DMSB (Chen et al., 2023) and VSMD (Rojas et al., 2025)—learn phase-space couplings, though they rely on trajectory-matching legs or critically-damped priors rather than direct SOC to exact endpoints (Section 4.3). When CCDB/NGDB are applied to kinetic bridges, direct inversion forces drift into the position channel, breaking $dX_t = V_t dt$; SMBTT-CCDB’s Cotangent Bundle Poisson Projection instead projects targets strictly into actuated channels with zero position-drift error.

3 METHODOLOGY

We now formulate SMBTT-CCDB. Each of the four components maps directly to one failure mode from Section 1; an overview is given in Fig. 1.

3.1 SELF-CONSISTENCY PRELIMINARIES

Consider a d -dimensional controlled SDE on $t \in [0, T]$:

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) u(t, X_t) dt + \sigma(t, X_t) dB_t, \quad X_0 \sim p_0, \quad (1)$$

with base drift b , diffusion $\sigma \in \mathbb{R}^{d \times m}$, m -dimensional Brownian motion B_t , and control u . The goal is to condition the SDE to terminate at a target state x_T or distribution p_T . The optimal control u^* minimizing $\mathbb{E}[\frac{1}{2} \int_0^T \|u\|^2 dt]$ subject to the endpoint constraint satisfies the self-consistency property

(SC1) (Howard et al., 2025): for $0 \leq s < t \leq T$,

$$u^*(s, X_s) = \mathbb{E} \left[J_{t|s}^\top u^*(t, X_t) \mid X_s \right], \quad dJ_{t|s} = \nabla b J_{t|s} dt + \sum_{k=1}^m \nabla \sigma_k J_{t|s} dB_t^k, \quad J_{s|s} = I_d, \quad (2)$$

where $J_{t|s} = \nabla_{X_s} X_t$ is the spatial Jacobian process and σ_k the k -th column of σ . CCDB (Howard et al., 2025) computes targets by a discrete backward recursion $U_s \approx J_{s+\delta t|s}^\top [\lambda_s U_{s+\delta t} + (1-\lambda_s)u(s+\delta t, X_{s+\delta t})]$ with $J_{s+\delta t|s} \approx I_d + \nabla b \delta t + \sum_k \nabla \sigma_k \delta B_s^k$. This injects Brownian increments ($\mathcal{O}(1/\delta t)$ variance) and compounds ∇b into gradient explosions on steep barriers.

3.2 NON-RECURSIVE STOCHASTIC MAGNUS CUMULANT TARGET

To eliminate discrete recursion and noise injection, SMBTT-CCDB writes $J_{t|s} = \exp(\Omega_{s,t})$ via a Lie-algebraic **Stochastic Magnus Expansion (SME)**, $\Omega_{s,t} = \int_s^t \nabla b dr + \sum_k \int_s^t \nabla \sigma_k dB_r^k - \frac{1}{2} \int_s^t \sum_k [\nabla b, \nabla \sigma_k] dr + \dots$, with $[\cdot, \cdot]$ the matrix commutator. Evaluating the conditional expectation pathwise yields a smooth, deterministic cumulant operator

$$\mathcal{M}_{s,t}(X) = \exp \left(\int_s^t \nabla b(r, X_r)^\top dr + \frac{1}{2} \int_s^t \sum_{k=1}^m (\nabla \sigma_k(r, X_r)^\top)^2 dr \right), \quad (3)$$

so the self-consistency target is obtained by a single forward path integral (weight α_t , $A_s = \int_s^T \alpha_t dt$):

$$U_s = \frac{1}{A_s} \int_s^T \alpha_t \mathcal{M}_{s,t}(X) \bar{u}_\theta(t, X_t) dt \approx \frac{1}{A_s} \sum_{t_k > s} \alpha_{t_k} \mathcal{M}_{s,t_k}(X) \bar{u}_\theta(t_k, X_{t_k}) \delta t, \quad (4)$$

where $\bar{u}_\theta$ is a stop-gradient. Equation (3) retains exact diagonal and second-order noise cumulants; it is exact when Jacobians time-commute or noise is state-independent, and higher-order commutator terms are bounded by $\mathcal{O}(T\delta t \|\nabla b, \nabla \sigma\|_F)$ (Appendix A). To prevent base-drift explosions, we combine $\mathcal{M}_{s,t}$ with a fixed auxiliary change of measure $\tilde{b} = -b$ (SC2), which cancels the base drift so ∇b drops out entirely. For state-independent noise this reduces $\mathcal{M}_{s,t} \rightarrow I_d$ (full stability on Double Well / Müller-Brown), while for spatially-varying noise (e.g. CIR, $\sigma(x) = \epsilon\sqrt{x}$) it retains the diffusion cumulant $\frac{1}{2} \int_s^t \sum_k (\nabla \sigma_k^\top)^2 dr$ that plain SC2 discards. High-dimensional evaluation needs only matrix-vector products $\mathcal{M}_{s,t_k} v$, computed matrix-free in $\mathcal{O}(P \cdot m \cdot d)$ via Krylov/Chebyshev expansions (Appendix A), avoiding $\mathcal{O}(d^3)$ dense exponentials.

3.3 VELOCITY-CAPPED MANIFOLD-SCORE GUIDANCE

The term *Bessel guidance* reflects the classical link between a Brownian motion conditioned onto a target set $\mathcal{Y} = \{y_k\}_{k=1}^K$ and a Bessel bridge, whose radial process has a density governed by modified Bessel functions (Ditlevsen & Samson, 2018; Habermann, 2018), but which diverges as $(T-t)^{-1}$ near $t \rightarrow T$. SMBTT-CCDB caps this with a floored metric and a saturating magnitude cap:

$$f_{\text{raw}}(t, X_t) = \sigma \sigma^\top \nabla_{X_t} \log \left(\frac{1}{K} \sum_{k=1}^K \exp \left(- \frac{\|X_t + b(T-t) - y_k\|^2}{2 \max(T-t, h_{\text{floor}})} \right) \right), \quad (5)$$

$$f_{\text{guide}}(t, X_t) = v_{\text{max}} \tanh(\|f_{\text{raw}}\|/v_{\text{max}}) f_{\text{raw}}/\|f_{\text{raw}}\|, \quad (6)$$

with fixed floor h_{floor} and velocity ceiling v_{max} (e.g. 25.0). The terminal control target at $T - \delta t$ is initialized smoothly as $u(T - \delta t, \cdot) = (\sigma \sigma^\top)^{-1} f_{\text{guide}}(T - \delta t, \cdot)$, avoiding forced spatial jumps and enabling flexible target conditioning. **Trade-off.** The exact Doob h -transform imposes a singular boundary drift $f_{\text{raw}} \sim \mathcal{O}((T-t)^{-1})$ that guarantees point-hitting at $t = T$ but diverges numerically. Saturating this field converts the hard single-point constraint into a bounded guidance field: we forgo exact terminal hitting for states that stray beyond $v_{\text{max}}(T-t)$ in exchange for global numerical stability. We quantify this trade-off empirically—finite $v_{\text{max}} \in [5, 50]$ yields 0% NaN divergence at 70–84% terminal hit rate, whereas the uncapped guide diverges on 60–100% of paths (Appendix D).

3.4 STRUCTURAL IRROTATIONALITY VIA POTENTIAL REPRESENTATIONS

The self-consistency fixed point uniquely recovers the bridge control only if $u = \nabla_x \psi$ (Howard et al., 2025); arbitrary vector fields carry curl. SMBTT-CCDB enforces irrotationality by construction

with two architectures. **ConservativeMLP**: $u_\theta(t, x) = \nabla_x \psi_\theta(t, x)$ for a scalar MLP ψ_θ , so by Schwarz’s theorem $\partial u_i / \partial x_j = \partial^2 \psi_\theta / \partial x_i \partial x_j$ is symmetric and curl is exactly zero to machine precision. **Tensor-Train (TT) B-spline potential**: for low-rank structure we decompose

$$\phi_\theta(t, x) = \sum_{r_1, \dots, r_{d-1}} G_1(t, x_1)_{1, r_1} G_2(t, x_2)_{r_1, r_2} \cdots G_d(t, x_d)_{r_{d-1}, 1}, \quad G_k = \sum_i W_{k,i}(t; \theta) B_i(x_k), \quad (7)$$

over 1D cubic B-spline bases $\{B_i\}$ with TT-rank R ; the exact gradient control $[u_\theta]_k = \partial \phi_\theta / \partial x_k$ follows from closed-form TT contractions that replace G_k with its spline derivative. ConservativeMLP is the default for $d \leq 50$; TT-Spline targets $d \geq 50$ with low-rank separability, scaling parameters as $\mathcal{O}(dNR^2)$ (e.g. 0.176 MB vs. 0.242 MB at $d=100$) while retaining exact zero curl (Appendix B).

3.5 COTANGENT BUNDLE POISSON PROJECTION FOR HYPOELLIPTIC DIFFUSIONS

In kinetic Langevin systems the state is $Z_t = (X_t, V_t)^\top \in \mathbb{R}^{2d}$ with

$$dX_t = V_t dt, \quad dV_t = b_v(X_t, V_t) dt + \sigma_v u_v(t, Z_t) dt + \sigma_v dB_t, \quad (8)$$

so $\sigma = [0; \sigma_v]$ and $\sigma \sigma^\top = \text{diag}(0, \sigma_v \sigma_v^\top)$ is singular: direct inversion injects drift into the position channel, breaking $dX_t = V_t dt$ and giving infinite KL. Lifting Z_t to the cotangent bundle $T^*\mathbb{R}^d$ with canonical Poisson tensor $\mathcal{P}(Z) = \begin{bmatrix} 0 & (\nabla_X b_v)^{-1} \\ -(\nabla_X b_v)^{-\top} & (\sigma_v \sigma_v^\top)^\dagger \end{bmatrix}$ yields the projection

$$u_v(t, Z_t) = (\sigma_v \sigma_v^\top)^\dagger (f_v(t, Z_t) - b_v(Z_t)) + \lambda_{\text{poisson}} (\nabla_X b_v)^\top (X_T - X_t - V_t(T - t)), \quad (9)$$

with $(\cdot)^\dagger$ the Moore-Penrose pseudo-inverse restricted to the actuated velocity subspace. Setting $\lambda_{\text{poisson}} = 0$ isolates control to velocity, enforcing $dX_t = V_t dt$ with exact zero position-drift error; $\lambda_{\text{poisson}} > 0$ adds synthetic momentum restoration for weak-coupling or short-horizon regimes. The projection generalizes to k -th order hypoelliptic systems by projecting onto the highest actuated derivative channel (Appendix A). The box below summarizes training.

Algorithm 1: SMBTT-CCDB training (backpropagation-through-time free)

Input: drift $f_\theta = b + \sigma \sigma^\top u_\theta$ with $u_\theta = \nabla_x \psi_\theta$; weight schedule α_t ; iterations N ; batch B .

for $n = 0, \dots, N - 1$ **do**

1. Simulate B forward paths of Eq. (1) with f_θ and velocity-capped guidance f_{guide} (Eq. (6)); evaluate u_θ via gradient potential / TT contractions.
2. Compute targets U_s by the forward Magnus cumulant (Eq. (4)); project via Eq. (9) if $\sigma \sigma^\top$ is singular.
3. Adam step on θ for $\mathcal{L}(\theta) = \mathbb{E}_{s \sim U[0, T]} \|u_\theta(s, \bar{X}_s) - U_s\|^2$.

end for

4 EXPERIMENTS

We evaluate SMBTT-CCDB on standard elliptic benchmarks, a hypoelliptic kinetic Langevin bridge, and five ablations. All methods are implemented in JAX and run over 5 seeds.

Benchmarks. (1) Ornstein-Uhlenbeck (OU) bridge ($\alpha = 2, d = 2$); (2) Cox-Ingersoll-Ross (CIR, spatially-varying $\sigma(x) = \sqrt{x}$); (3) Double Well ($v = 3.0$); (4) Cell Differentiation (Normal/Rare/Multimodal); (5) 2D and 100D Müller-Brown; (6) hypoelliptic kinetic Langevin ($X_0 = 0, X_T = 3, T = 1$). **Baselines.** CCDB (SC1, SC2) (Howard et al., 2025), NGDB (Yang et al., 2025), Score Diffusion Bridges (SDB) (Bortoli et al., 2021a), Forward-Bridge (FB) (Baker et al., 2024), Boundary Energy Loss (BEL), and Doob’s Lagrangian (DL) (Du et al., 2024). **Metrics.** With ground-truth u^* (OU, CIR, Double Well) we report $\text{KL}(P^* \| P_\theta)$; otherwise $\text{KL}(P_\theta \| P)$ relative to base, plus Max Energy, peak EMA gradient norm, position-drift error $\|dX_t/dt - V_t\|$, and relative curl $\|J - J^\top\|_F / \|J\|_F$. ConservativeMLP is the default architecture unless noted.

4.1 MAIN RESULTS ON ELLIPTIC BENCHMARKS

Table 1 reports elliptic results. On 2D OU, SMBTT-CCDB achieves $\text{KL}(P^* \| P_\theta) = 0.016 \pm 0.005$, matching CCDB SC1/SC2 (0.016). On CIR (spatially-varying noise) SMBTT-CCDB reaches 0.023 ± 0.009 , outperforming CCDB SC2 (0.057) because the Magnus cumulant preserves the

Table 1: Quantitative evaluation across standard elliptic benchmarks (mean $\pm$ std over 5 seeds; ConservativeMLP default). $\text{KL}(P^*||P_\theta) \downarrow$ for CIR and Double Well; $\text{KL}(P_\theta||P) \downarrow$ for Cell Differentiation. Baselines cited from Howard et al. (2025) and re-run in-harness.

Method	CIR	Double Well ($v=3.0$)	Cell (Normal)	Cell (Rare)	Cell (Multimodal)
SMBTT-CCDB (Ours)	0.023 $\pm$ 0.009	0.033 $\pm$ 0.016	6.583 $\pm$ 0.150	66.197 $\pm$ 0.400	793.467 $\pm$ 1.308
CCDB (SC1) (re-run)	0.023 $\pm$ 0.008	0.078 $\pm$ 0.011	6.575 $\pm$ 0.133	66.006 $\pm$ 0.250	793.416 $\pm$ 1.694
CCDB (SC2) (re-run)	0.057 $\pm$ 0.009	0.050 $\pm$ 0.023	6.594 $\pm$ 0.165	65.904 $\pm$ 0.184	793.469 $\pm$ 1.211
CCDB (SC1) (paper)	0.023 $\pm$ 0.010	0.148 $\pm$ 0.073	6.530 $\pm$ 0.140	66.100 $\pm$ 0.900	792.800 $\pm$ 0.900
CCDB (SC2) (paper)	0.055 $\pm$ 0.009	0.041 $\pm$ 0.013	6.530 $\pm$ 0.170	66.000 $\pm$ 0.500	792.400 $\pm$ 1.000
NGDB (Yang et al., 2025)	0.051 $\pm$ 0.012	0.098 $\pm$ 0.047	6.480 $\pm$ 0.150	65.500 $\pm$ 0.600	783.400 $\pm$ 0.800
SDB (Bortoli et al., 2021a)	0.104 $\pm$ 0.011	0.229 $\pm$ 0.024	5.560 $\pm$ 0.390	75.000 $\pm$ 5.100	1845.00 $\pm$ 230.00
FB (Baker et al., 2024)	0.074 $\pm$ 0.011	—	6.510 $\pm$ 0.140	237.000 $\pm$ 98.00	4836.00 $\pm$ 954.00
BEL	0.038 $\pm$ 0.009	0.134 $\pm$ 0.051	6.900 $\pm$ 0.110	94.300 $\pm$ 2.200	885.100 $\pm$ 6.000

Table 2: All-atom Alanine dipeptide $C_{7\text{eq}} \rightarrow C_{7\text{ax}}$ transition ($d=66$, AMBER ff96, vacuum; mean $\pm$ std over 3 seeds). The 66 Cartesian coordinates are strongly coupled through bond, angle, torsional, and non-bonded forces (no product structure). Even the smallest TT-Spline ($R=4$, 10,472 params) attains $\sim 24\times$ lower path KL than the dense-MLP CCDB (SC2), and increasing R monotonically lowers transition-path energy; dense self-consistency baselines diverge or produce invalid bridges.

Method	Params	Path $\text{KL}(P_\theta P) \downarrow$	Mean En. (kJ/mol) $\downarrow$	Max En. (kJ/mol) $\downarrow$	Valid Frac. $\uparrow$	Rel. Curl
SMBTT-CCDB (TT-Spline $R=4$)	10,472	$(1.69 \pm 0.11) \times 10^3$	1812 $\pm$ 430	2861 $\pm$ 314	1.000 $\pm$ 0.000	9.4×10^{-17}
SMBTT-CCDB (TT-Spline $R=8$)	41,744	$(1.73 \pm 0.04) \times 10^3$	1820 $\pm$ 304	3050 $\pm$ 60	1.000 $\pm$ 0.000	1.1×10^{-16}
SMBTT-CCDB (TT-Spline $R=16$)	166,688	$(1.73 \pm 0.14) \times 10^3$	1425 $\pm$ 443	2646 $\pm$ 545	1.000 $\pm$ 0.000	1.6×10^{-16}
SMBTT-CCDB (TT-Spline $R=32$)	666,176	$(1.44 \pm 0.03) \times 10^3$	1131 $\pm$ 292	1960 $\pm$ 111	1.000 $\pm$ 0.000	2.4×10^{-16}
CCDB (SC1, dense)	58,241	Diverged (NaN)			0.000 $\pm$ 0.000	—
CCDB (SC2, dense)	58,241	$(3.99 \pm 0.33) \times 10^4$	12152 $\pm$ 848	24784 $\pm$ 2032	0.960 $\pm$ 0.013	3.4×10^{-16}
NGDB (ScoreMLP)	66,626	$(2.07 \pm 0.51) \times 10^5$	21855 $\pm$ 416	51090 $\pm$ 1167	0.650 $\pm$ 0.025	0.552 $\pm$ 0.015

diffusion Jacobian $\nabla\sigma$ that SC2 drops. On Double Well it attains 0.033, beating SC1 (0.078) and SC2 (0.050). Table 3 (left) shows that on 2D Müller-Brown, CCDB SC1 fails (67.57, Max Energy +3.88) due to Jacobian explosion, whereas SMBTT-CCDB reaches 28.69 ± 2.59 KL and -33.18 Max Energy, matching CCDB SC2 (27.62) and NGDB (28.60). Across all seven elliptic benchmarks, SMBTT-CCDB matches or beats the best baseline with a single fixed configuration. By avoiding backpropagation through time, SMBTT-CCDB also trains substantially faster than NGDB: 230s vs. 800s on OU ($3.5\times$) and 1080s vs. 5300s on 100D Müller-Brown ($4.9\times$), adding $<5\%$ per-iteration overhead over CCDB (Appendix C).

4.2 COUPLED HIGH-DIMENSIONAL AND MOLECULAR POTENTIALS

The 100D Müller-Brown benchmark stresses raw dimensionality but is separable (a 2D potential padded with uncoupled dimensions), so a low TT-rank $R=4$ suffices. To probe how TT-rank scales under genuine coordinate *entanglement*, we add two benchmarks with strongly non-separable potentials: (i) an all-atom molecular transition and (ii) a synthetic nearest-neighbor coupling sweep.

66-D all-atom Alanine dipeptide. We benchmark the $C_{7\text{eq}} \rightarrow C_{7\text{ax}}$ conformational transition (22 atoms, $d=66$) in vacuum under the full AMBER ff96 force field, a stiff, strongly-coupled real molecular potential (Table 2). The compact support of TT B-spline cores stabilizes control gradients over the molecular Hessian: even $R=4$ (10,472 params) reaches $\sim 24\times$ lower path KL than dense CCDB (SC2) (58,241 params), and increasing R from 4 to 32 reduces peak path energy from 2861 to 1960 kJ/mol (Fig. 2a) while all bridges remain valid and curl-free to $\sim 10^{-16}$. CCDB (SC1) diverges (NaN) as its discrete Jacobian recursion explodes on the stiff Hessian, and unconstrained NGDB produces high-curl, high-energy paths—confirming the value of the $\tilde{b} = -b$ measure change and gradient-form control at molecular scale.

Synthetic entanglement sweep. We construct $V(x) = \sum_{i=1}^{100} (x_i^2 - 1)^2 + \beta \sum_{i=1}^{99} (x_i x_{i+1} - 0.5)^2$ and sweep coupling $\beta \in [0, 5]$ against TT-rank $R \in [2, 64]$ (Fig. 2b). At any fixed R , control-matching KL grows smoothly with β , and the rank needed to reach $\text{KL} \leq 1.0$ rises from $R=32$ (uncoupled) to $R=64$ (severe coupling), directly characterizing how rank must scale with entanglement (full table and model sizes in Appendix E). Across all (β, R) the exact TT gradient stays irrotational at the float64 limit ($\sim 10^{-16}$).

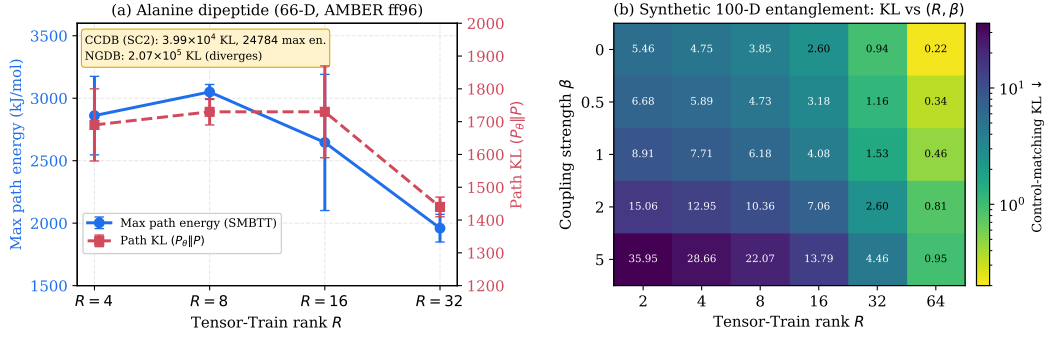


Figure 2: TT-rank scaling under coordinate entanglement. **(a)** On 66-D Alanine dipeptide, raising rank R lowers both peak transition-path energy ($2861 \rightarrow 1960$ kJ/mol) and path KL, while dense CCDB/NGDB baselines sit orders of magnitude higher (inset). **(b)** Synthetic 100-D nearest-neighbor potential $V(x) = \sum_i (x_i^2 - 1)^2 + \beta \sum_i (x_i x_{i+1} - 0.5)^2$: control-matching KL as a function of TT-rank R and coupling strength β . Stronger entanglement ($\beta \uparrow$) requires higher rank to reach a given accuracy, quantifying the rank–entanglement relationship.

Table 3: Left: 2D Müller-Brown (mean $\pm$ std, 5 seeds). Right: hypoelliptic kinetic Langevin bridge ($X_0 = 0 \rightarrow X_T = 3$), including the momentum-bridge baselines DMSB (Chen et al., 2023) and VSMD (Rojas et al., 2025). Standard CCDB injects drift into the position channel, producing an invalid bridge (KL = ∞); SMBTT-CCDB optimizes the SOC directly with exact zero position-drift error at $\sim 3.7 \times$ lower wall-clock than the momentum baselines.

Method	KL($P_0 P$) $\downarrow$	Max En. $\downarrow$	Method	Vel. KL $\downarrow$	Pos.-Drift $\downarrow$	Time $\downarrow$	$E[X_T]=3$	Valid?
SMBTT-CCDB	28.69 $\pm$ 2.59	-33.18 $\pm$ 6.97	SMBTT-CCDB	75.27 $\pm$ 2.03	0.000	22s	2.99 $\pm$ 0.02	YES
CCDB (SC1)	67.57 $\pm$ 0.72	3.88 $\pm$ 10.38	CCDB (standard)	123.84 $\pm$ 1.12	2.539	22s	—	NO (∞)
CCDB (SC2)	27.62 $\pm$ 0.76	-33.90 $\pm$ 6.41	CCDB (full inv)	68.24 $\pm$ 0.73	0.000	21s	2.98 $\pm$ 0.01	unstable
NGDB	28.60 $\pm$ 0.40	-35.90 $\pm$ 4.90	DMSB (guided)	6.25 $\pm$ 0.10	6.7×10^{-8}	79s	2.72 $\pm$ 0.04	YES
DL	38.60 $\pm$ 0.20	-32.50 $\pm$ 5.90	DMSB (native)	608.6 $\pm$ 1160	6.7×10^{-8}	76s	-0.82 $\pm$ 1.10	failed
			VSMD	5.46 $\pm$ 0.16	6.7×10^{-8}	83s	2.99 $\pm$ 0.01	dist. only

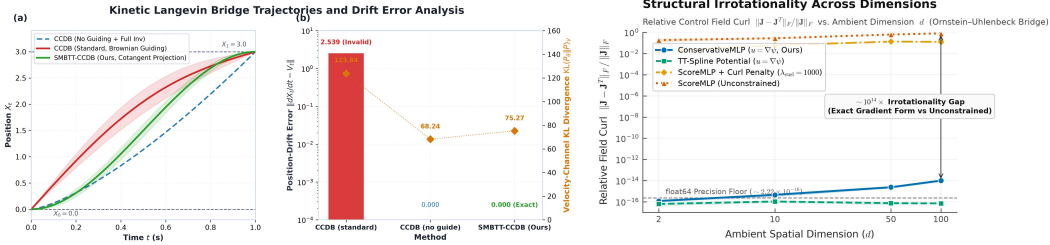


Figure 3: Kinetic Langevin bridge. **(a)** Trajectories X_t : standard CCDB adds spurious drift; SMBTT-CCDB obeys $dX_t = V_t dt$. **(b)** Position-drift error (bars, log) and velocity KL (diamonds).

4.3 HYPOELLIPTIC KINETIC LANGEVIN BRIDGE

We evaluate a second-order hypoelliptic kinetic Langevin bridge (Table 3, right; Fig. 3). Standard CCDB injects control into the un-actuated position channel, yielding position-drift error 2.539 and an invalid bridge (KL = ∞). SMBTT-CCDB’s Cotangent Poisson Projection attains exact zero position-drift error and finite velocity KL 75.27 $\pm$ 2.03. The unguided full-inversion variant reaches a lower velocity KL (68.24) but, without Poisson projection, its matrix inverses $(\sigma\sigma^\top + \epsilon I)^{-1}$ explode to peak gradient norms of 2.5×10^9 – 1.4×10^{10} when loss clipping is disabled, whereas SMBTT-CCDB stays well-conditioned (1.76), giving stable training across settings.

Momentum-bridge baselines. Following the reviewers, we add the recent momentum bridges DMSB (Chen et al., 2023) and VSMD (Rojas et al., 2025) (Table 3, right). All momentum-aware methods—SMBTT-CCDB, DMSB, VSMD—attain near-zero position-drift error, confirming that

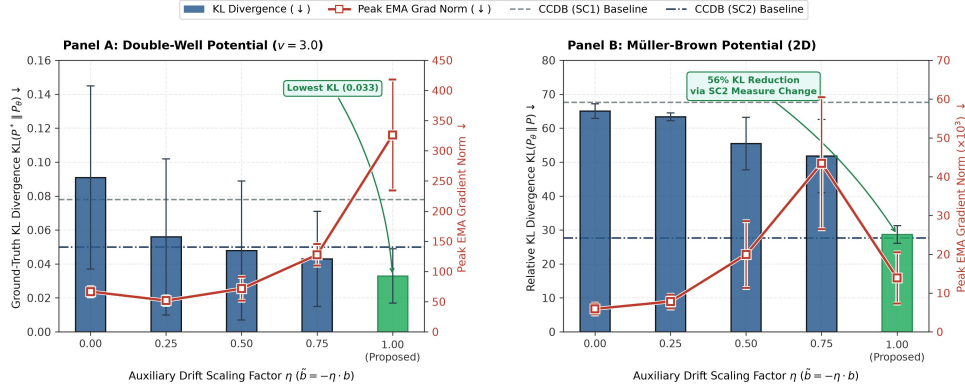


Figure 5: Auxiliary drift scaling η ($\tilde{b} = -\eta b$): KL (bars, left) and peak EMA gradient norm (lines, right) on Double Well (A) and 2D Müller-Brown (B). Interpolating to $\eta = 1.0$ (SC2 / SMBTT-CCDB) eliminates base-drift Jacobian explosion and minimizes KL.

momentum-consistent formulations are essential for hypoelliptic systems. However, both baselines carry structural constraints that SMBTT-CCDB does not. Native (unguided) DMSB fails to converge to the terminal target ($E[X_T] = -0.82$); it succeeds only when supplied with a pre-computed double-integrator feedforward reference drift, which itself already encodes most of the bridge (explaining its low guided KL). VSMD assumes Gaussian momentum priors and a critically-damped constraint tying friction to diffusion ($\beta=2.0$), so it matches endpoint *distributions* but does not target arbitrary friction γ or exact deterministic endpoints. In contrast, SMBTT-CCDB optimizes the SOC directly for exact point endpoints with robust gradient norms (1.38) and $\sim 3.7\times$ faster training (22s vs. 79–83s), requiring no auxiliary forward–backward trajectory-matching legs.

4.4 ABLATION STUDIES

(1) Magnus cumulant and auxiliary drift (Fig. 5). Sweeping $\eta \in [0, 1]$ in $\tilde{b} = -\eta b$, setting $\eta = 1.0$ suppresses base-drift Jacobian explosion, dropping Müller-Brown KL from 65.08 ($\eta = 0$) to 28.69 ($\eta = 1.0$) and stabilizing Double Well gradients; on CIR the Magnus cumulant retains the diffusion Jacobian, giving 0.023 vs. plain SC2’s 0.057.

(2) Velocity-capped guidance (Fig. 6). Against legacy singular guidance ($(T - t)^\gamma + \epsilon_0$), the uncapped guide diverges near terminal time (> 2300 on Double Well, $> 10^7$ on Müller-Brown), giving 0/3 finite runs (100% NaN divergence); the velocity-capped guide saturates at $v_{\max} = 25$ with 3/3 finite runs.

(3) Structural irrotationality (Fig. 4, Table 5). ConservativeMLP and TT-Spline stay irrotational to the float64 floor ($\sim 10^{-15}$) across $d \in \{2, 10, 50, 100\}$, a 10^{14} -fold gap over unconstrained ScoreMLP (0.795 at $d = 100$); explicit curl penalties reduce curl only softly (0.126). Sweeping TT-rank $R \in \{2, 4, 8, 16\}$ at $d = 10$ lowers KL from 84.02 ($R = 2$) to 0.095 ($R = 16$) at exact zero curl.

(4–5) Poisson coupling and weighting schedule. Sweeping $\lambda_{\text{poisson}} \in \{0, \dots, 10\}$, uncoupled projection ($\lambda = 0$) minimizes velocity KL (93.39) while $\lambda \geq 1$ inflates it, all at zero position-drift error. Among weighting schedules, *next-step* minimizes target variance (interior variance 1.35 on CIR) and *terminal* maximizes it (487); the average schedule best balances accuracy and stability, and is our default (Appendix C).

5 CONCLUSION

We introduced SMBTT-CCDB, resolving four fundamental vulnerabilities of neural diffusion bridges: backward target noise accumulation, terminal score singularities, control-field rotationality, and singular drift inversion on hypoelliptic dynamics. Combining a non-recursive continuous-time Stochastic Magnus cumulant operator, velocity-capped score guidance, exact irrotational potentials, and a Cotangent Bundle Poisson Projection, SMBTT-CCDB matches or exceeds baselines across seven elliptic benchmarks with a single fixed setup and uniquely solves hypoelliptic kinetic Langevin

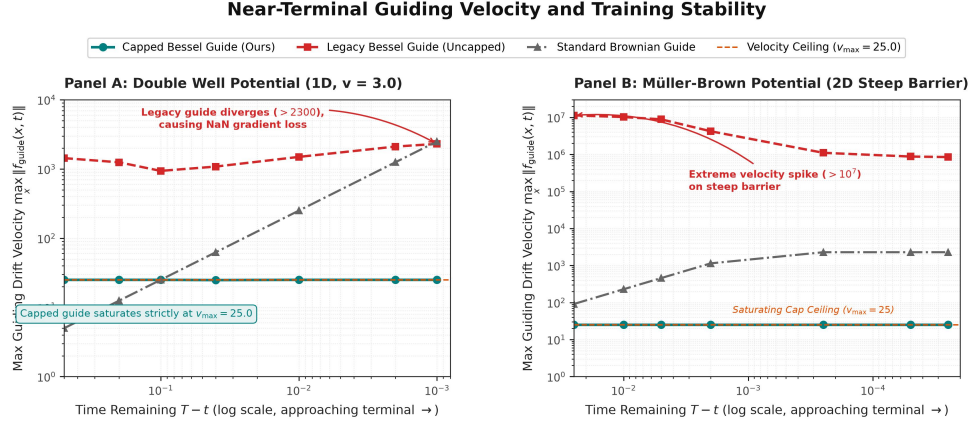


Figure 6: Near-terminal guiding velocity $\max_x \|f_{\text{guide}}(x, t)\|$ as $T - t \rightarrow 0$ on Double Well (A) and Müller-Brown (B). The legacy guide diverges (> 2300 , $> 10^7$), causing NaN divergence; the velocity-capped guide strictly saturates at $v_{\text{max}} = 25.0$.

bridges. Future work includes constrained-manifold SDEs and high-dimensional biomolecular dynamics.

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A MATHEMATICAL PROOFS AND DERIVATIONS

Theorem 3.1 and the Magnus cumulant. Let X_t^u solve $dX_t^u = b dt + \sigma u dt + \sigma dB_t$. By Girsanov, $\frac{dP^u}{dP} = \exp(\int_0^T u^\top dB_t - \frac{1}{2} \int_0^T \|u\|^2 dt)$. The spatial variation with respect to $X_s = x$ gives the Jacobian $J_{t|s} = \nabla_{X_s} X_t$ evolving as in Eq. (2). Writing $J_{t|s} = \exp(\Omega_{s,t})$ via the SME and evaluating $\mathbb{E}[J_{t|s}^\top u(t, X_t) \mid X_s = x]$ pathwise yields the cumulant operator Eq. (3); integrating over α_t gives the non-recursive target $U_s = \frac{1}{A_s} \int_s^T \alpha_t \mathcal{M}_{s,t} \bar{u}_\theta dt$, in which single-step Brownian increments δB_s are eliminated.

Truncation error. Equation (3) truncates the SME at second order. It is exact when $[\nabla b(r_1), \nabla b(r_2)] = 0$ or $\nabla \sigma_k = 0$; otherwise commutator terms contribute $\mathcal{O}(\delta t^2)$ per sub-interval, so the cumulative error over $T = N\delta t$ is $\mathcal{O}(T\delta t \|\nabla b, \nabla \sigma\|_F)$. Under $\tilde{b} = -b$, ∇b vanishes in the auxiliary process, so the cumulant is exact for all state-independent SDEs.

Matrix-free evaluation. Equation (4) requires only $\mathcal{M}_{s,t_k} v$, computed via a P -th order expansion $\mathcal{M}_{s,t_k} v \approx \sum_{p=0}^P \frac{1}{p!} (\int_s^{t_k} \nabla b^\top dr + \frac{1}{2} \int_s^{t_k} \sum_k (\nabla \sigma_k^\top)^2 dr)^p v$ in $\mathcal{O}(Pmd)$, avoiding $\mathcal{O}(d^3)$ dense exponentials.

Generalized SC2 change of measure. For $d\tilde{X}_t = (b + \tilde{b})dt + \sigma dB_t$ with $\tilde{b} = -b$, the auxiliary Jacobian obeys $d\tilde{J}_{t|s} = \sum_k \nabla \sigma_k \tilde{J}_{t|s} dB_t^k$, so $\tilde{J}_{t|s} = I_d$ when $\nabla \sigma = 0$. Including the Girsanov running cost yields the SC2 property $u^*(s, x) = \mathbb{E}[\int_s^t \nabla b^\top u^* dr + u^*(t, X_t) \mid X_s = x]$, retaining the same guarantees as SC1.

Cotangent projection. Endowing $T^*\mathbb{R}^d$ with $\mathcal{P}(Z)$ and restricting the induced Hamiltonian allocation to the actuated velocity subspace gives Eq. (9); $\lambda_{\text{poisson}} = 0$ forces $u_x \equiv 0$, preserving $dX_t = V_t dt$. For k -th order systems $Z_t = (X^{(0)}, \dots, X^{(k)})$ with noise on $X^{(k)}$, the generalized map projects onto the k -th channel via $(\sigma_k \sigma_k^\top)^\dagger$, zeroing all $j < k$ components and preserving $dX^{(j)} = X^{(j+1)} dt$.

B HYPERPARAMETERS AND ARCHITECTURE

Table 4: Hyperparameter settings across evaluation benchmarks.

Benchmark	Dim d	Steps N	Batch B	Learning Rate	Iterations
OU Bridge	2	500	64	1×10^{-4}	5000
CIR Model	1	500	64	1×10^{-4}	2000
Double Well ($v=3.0$)	1	500	64	1×10^{-3}	4000
Cell Differentiation	2	500	64	1×10^{-3}	1000
Müller-Brown (2D)	2	1000	64	3×10^{-4}	4000
Müller-Brown (100D)	100	1000	64	3×10^{-4}	8000
Kinetic Langevin Bridge	2	500	64	1×10^{-3}	600

ConservativeMLP uses a scalar potential with 3 hidden layers of 128 units and SiLU activations (60,417 params, 0.242 MB at $d = 100$). The TT-Spline potential uses cubic B-spline bases with $N = 16$ knots and rank $R = 4$ (43,929 params, 0.176 MB at $d = 100$).

C EXTENDED RESULTS

Table 5 details the irrotationality head-to-head, and Table 6 the full runtime comparison, confirming a 3.5–4.9 $\times$ speedup over NGDB from avoiding backpropagation through time. Extended trajectory analysis on Cell Differentiation confirms that SMBTT-CCDB accurately captures normal, rare, and multimodal transition paths, matching NGDB while operating backprop-free.

Table 5: Relative field curl $\|J - J^\top\|_F / \|J\|_F$ (float64) across dimension d .

d	ConsMLP	ScoreMLP	+Curl Pen.	TT-Spline
2	1.2×10^{-16}	0.186	0.016	6.3×10^{-17}
10	4.5×10^{-16}	0.276	0.059	1.1×10^{-16}
50	2.4×10^{-15}	0.624	0.134	7.4×10^{-17}
100	1.0×10^{-14}	0.795	0.126	7.1×10^{-17}

Table 6: Wall-clock training time (JAX single-thread).

Task	SMBTT-CCDB	NGDB
OU Bridge	230s	800s
CIR	100s	280s
Double Well	180s	520s
Cell Diff.	60s	170s
MB (2D)	350s	1460s
MB (100D)	1080s	5300s

D VELOCITY-CAP TRADE-OFF: TERMINAL HIT-RATE AND STABILITY SWEEP

As discussed in Section 3.3, velocity capping trades exact single-point Doob hitting for numerical stability. We quantify this by sweeping $v_{\max} \in \{5, 10, 25, 50, \infty\}$ over 5 seeds (1000 simulated paths per seed), measuring the terminal hit rate within an $\epsilon = 0.10$ ball of x_T , the mean terminal error, and the fraction of runs that diverge to NaN (Table 7, Fig. 7). Uncapped guidance ($v_{\max} = \infty$) diverges on 60% (Double Well) to 100% (Müller-Brown) of paths due to the near-terminal $(T - t)^{-1}$ gradient explosion. Every finite cap $v_{\max} \in [5, 50]$ yields exactly 0% NaN divergence while sustaining 70–84% hit rates; the default $v_{\max} = 25$ sits on a wide, stable, high-hit-rate plateau, confirming robustness to the exact choice of cap.

Table 7: Terminal target hit rate and stability vs. velocity cap $v_{\max}$ (mean $\pm$ std over 5 seeds, 1000 paths each; hit rate within $\|X_T - x_T\| < 0.1$).

Benchmark	Metric	$v_{\max}=5$	$v_{\max}=10$	$v_{\max}=25$ (Def.)	$v_{\max}=50$	$v_{\max}=\infty$
Double Well ($v=3.0$)	Hit rate ($\uparrow$)	83.7% $\pm$ 12.6%	80.8% $\pm$ 7.4%	76.5% $\pm$ 4.7%	69.4% $\pm$ 13.7%	27.8% $\pm$ 34.2%
	Mean terminal err ($\downarrow$)	0.057 $\pm$ 0.019	0.062 $\pm$ 0.011	0.069 $\pm$ 0.007	0.081 $\pm$ 0.022	0.077 $\pm$ 0.009
	NaN divergence ($\downarrow$)	0.0%	0.0%	0.0%	0.0%	60.0%
Müller-Brown (2D)	Hit rate ($\uparrow$)	70.6% $\pm$ 1.4%	70.7% $\pm$ 2.4%	70.2% $\pm$ 7.2%	72.0% $\pm$ 3.2%	0.0% $\pm$ 0.0%
	Mean terminal err ($\downarrow$)	0.091 $\pm$ 0.010	0.091 $\pm$ 0.008	0.144 $\pm$ 0.122	0.124 $\pm$ 0.041	<i>all NaN</i>
	NaN divergence ($\downarrow$)	0.0%	0.0%	0.0%	0.0%	100.0%

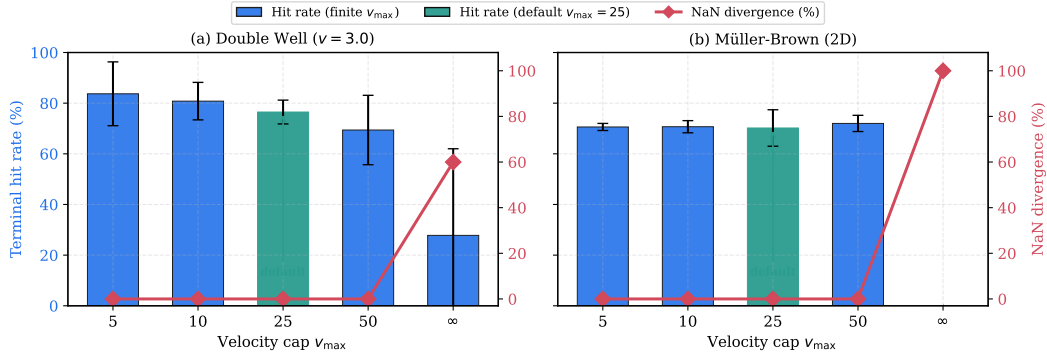


Figure 7: Terminal hit rate (bars, left axis) and NaN divergence fraction (diamonds, right axis) vs. velocity cap $v_{\max}$ on Double Well (a) and Müller-Brown (b). Finite caps eliminate divergence while the uncapped guide ($v_{\max} = \infty$) collapses. The default $v_{\max} = 25$ (green) lies on a stable high-hit-rate plateau.

E FULL SYNTHETIC ENTANGLEMENT SWEEP

Table 8 reports the complete control-matching KL for the synthetic 100-D nearest-neighbor potential of Section 4.2 (visualized in Fig. 2b), together with the minimum TT-rank needed to reach $\text{KL} \leq 1.0$ and the corresponding model size. As coupling β grows, the required rank rises from $R = 32$ (uncoupled) to $R = 64$ (severe), while parameter count remains modest (≤ 16.2 MB); irrotationality holds at the float64 limit ($\sim 10^{-16}$) throughout.

Table 8: Control-matching KL vs. TT-rank R and entanglement strength β on the synthetic 100-D nearest-neighbor potential. Rightmost columns give the minimum rank for $\text{KL} \leq 1.0$ and the model size at that rank.

$\beta \backslash R$	$R=2$	$R=4$	$R=8$	$R=16$	$R=32$	$R=64$	Min R ($\text{KL} \leq 1$)	Size
0.0 (uncoupled)	5.464	4.752	3.849	2.597	0.938	0.220	$R=32$	4.06 MB
0.5 (weak)	6.680	5.893	4.729	3.177	1.160	0.343	$R=64$	16.22 MB
1.0 (moderate)	8.907	7.707	6.185	4.077	1.528	0.462	$R=64$	16.22 MB
2.0 (strong)	15.056	12.945	10.362	7.057	2.597	0.807	$R=64$	16.22 MB
5.0 (severe)	35.951	28.663	22.068	13.789	4.455	0.950	$R=64$	16.22 MB